Exercise Sheet 13

The notation is that of Lecture 13. For $\mathfrak{sl}_3$ we use $h_1 = E_{11} - E_{22}$, $h_2 = E_{22} - E_{33}$, $e_1 = E_{12}$, $e_2 = E_{23}$, $f_1 = E_{21}$, $f_2 = E_{32}$ and $f_3 = E_{31} = [f_2, f_1]$. A weight μ is recorded by $(\mu(h_1), \mu(h_2))$, so $\alpha_1 = (2, -1)$ and $\alpha_2 = (-1, 2)$. Exercises 13.1–13.3 rebuild the lecture's argument in small cases. Exercise 13.4 shows why λ must be dominant integral. Exercises 13.5–13.6 apply the method to the adjoint module of $\mathfrak{sl}_3$ and to a sum of two simple ideals.

Exercise 13.1 (Rank one). Let $\mathfrak{g} = \mathfrak{sl}_2$ and $m \in \mathbb{Z}_{\geq 0}$. Write $v_0 = v_{m\omega_1}$ for the generator of $M(m\omega_1)$. By Lecture 12, the vectors $f^k v_0$ with $k \geq 0$ form a basis of $M(m\omega_1)$.

(i) Show that $U(\mathfrak{sl}_2) f^{m+1} v_0 = \operatorname{span}\{f^k v_0 : k \geq m+1\}$.

(ii) Let u_k be the image of $f^k v_0$ in $V(m\omega_1)$. Show that $u_0, \dots, u_m$ is a basis, and compute hu_k, fu_k and eu_k.

(iii) Prove directly that $V(m\omega_1)$ is simple, and identify it.

Exercise 13.2 (Reconstructing local nilpotence).

(i) Verify Lemma 13.4 of the lecture for $N = 2$ by expanding the right side.

(ii) In $\mathfrak{sl}_3$, show that $(\operatorname{ad} f_1)^2 \neq 0$ and $(\operatorname{ad} f_1)^3 = 0$.

(iii) Let V be any $\mathfrak{sl}_3$-module and $w \in V$ with $f_1^k w = 0$. Show that $f_1^{k+2} x w = 0$ for every $x \in \mathfrak{sl}_3$.

(iv) In the proof of Proposition 13.5, the case $y = e_i$ needs no defining relation of $V(\lambda)$. Why does the case $y = f_i$ need one?

Exercise 13.3 (Finiteness in coordinates). Let $\mathfrak{g} = \mathfrak{sl}_3$ and $\lambda = 2\omega_1 = (2, 0)$.

(i) Find all dominant weights $\mu \leq \lambda$.

(ii) List the weights of $V(\lambda)$. Show that each has multiplicity one, so $\dim V(\lambda) = 6$.

(iii) Let $\mathfrak{g}$ act on $\operatorname{Sym}^2(\mathbb{C}^3)$ by derivations, $x(ab) = (xa)b + a(xb)$. Show that b_1^2 satisfies the relations of $V(\lambda)$ and generates. Deduce $V(\lambda) \cong \operatorname{Sym}^2(\mathbb{C}^3)$.

Exercise 13.4 (Why dominant integrality is necessary).

(i) Let $\lambda \in \mathfrak{h}^*$ with $c = \lambda(h_i) \notin \mathbb{Z}_{\geq 0}$ for some i. Prove that $L(\lambda)$ is infinite-dimensional.

(ii) Let $\lambda \in P_+$ and fix i. Put $N' = \sum_{j \neq i} U(\mathfrak{g}) f_j^{m_j+1} v_\lambda$. Prove that $M(\lambda)/N'$ is infinite-dimensional.

(iii) Identify the steps of the lecture that use integrality and those that use nonnegativity of the m_i.

Exercise 13.5 (The adjoint module of $\mathfrak{sl}_3$). Let $\theta = \omega_1 + \omega_2 = (1, 1)$.

(i) Show that the dominant weights $\mu \leq \theta$ are θ and 0.

(ii) Show that $\dim V(\theta) \leq 8$, using $\dim M(\theta)_0$.

(iii) Using that $\mathfrak{sl}_3$ is simple, show that the adjoint module is $L(\theta)$. Deduce $\dim V(\theta)_0 = 2$.

Exercise 13.6 (Two simple ideals). Let $\mathfrak{g} = \mathfrak{sl}_2 \oplus \mathfrak{sl}_2$ act on $M_2(\mathbb{C})$ by

$$(X, Y) \cdot A = XA - AY.$$

(i) Verify that this is a representation.
(ii) Use the Cartan subalgebra spanned by $(h, 0)$ and $(0, h)$. Find the weight of each E_{ij}.
(iii) Show that $M_2(\mathbb{C}) \cong L(1, 1) \cong V_1 \boxtimes V_1$.
(iv) Restrict to the diagonal subalgebra $\{(X, X)\} \cong \mathfrak{sl}_2$ and decompose.